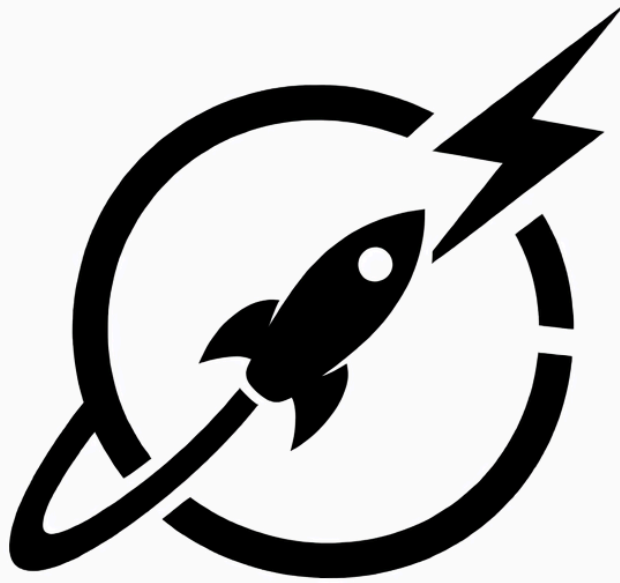


Critical Design Review

Electric dysfunction

Arcs Team 1



**ELECTRIC
DYSFUNCTION**

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Team structure

The Team consists of 14 members which have been split into 2 sub-teams; structures and avionics. Team Lead and Vice Team lead were decided through a vote at the start of the project. Once the 2 sub-teams had been decided, each team individually voted for a team lead to over-see each side of the Rocket's design process. All team-leads can communicate with each other to gauge the pace of the project, identify new tasks, discover any problems and assign work to other members of the team. Additionally, there is now a vice structures lead to help with better communication and organisation within the structures sub-team.

Name	Role	Lead	Assignment
Sam Orlin	Avionics	Avionics team lead	Hardware
Finn Woolley	Avionics		Data processing
Alistair Wye	Avionics		Software
Charlie sims	Avionics		Avionics bay & Integration
Olivia Gransden	Structures	Team lead	Organisation
Lara Burke	Structures	Vice team lead	Communications
Louis Goodwin	Structures	Structures team lead	Open Rocket design
Lewis Cairns	Structures		CAD design, Materials Research
Sophie Shandley	Structures		Research
Dan Hubbard	Structures	Vice structures lead	Open Rocket design
Robert Seddon	Structures		Research
Freddie Kendall	Structures		Simulations
Tyler Cain	Structures		Research
James Barnet	Structures		Research

Aims and objectives

An initial set of objectives was provided that the rocket must meet in terms of its structural aspects, avionics, and presentation. The main aim of our design was to design a rocket that would successfully launch with an impact speed between 7 and 10 metres per second, with a successful recovery system (the rocket should return in a re-launchable state), and a target altitude of 1900-2000ft. In addition to this, the payload should be at least 150 grams, collect at least 4 different types of data and should be able to successfully power itself for 40 minutes.

Budget and finance

Initially, each team member contributed £10 towards our equipment costs (£140 in total); however our current budget listed below exceeds this. The team is currently discussing a potential sponsorship to cover some of the costing, however this is yet to be confirmed. Alternatively, an additional contribution of approximately £5 from each team member will support the remainder of the finances. The motor (G78-7) that was selected was funded by the ARCS competition, therefore there was no cost attached to it on our behalf as well as the nose cone being 3D printed in the IForge so there is no cost attached. The diagram below shows the current progress of purchasing the different aspects of the rocket, separated into structures and avionics. The current cost of the rocket is approximately £187 including any shipping and VAT charges.

Description	Team	Quantity	Unit cost	Shipping	Total cost	link	status
MicroSD Module	Avionics	1	3.4	3.9	7.3	https://thepihut.com/products/adafruit-32gb-microsd-card	Recieved
Microcontroller	Avionics	1	19.8	0	19.8	https://thepihut.com/products/arduino-pro-mini	Recieved
Barometer	Avionics	1	10.6	0	10.6	https://thepihut.com/products/adafruit-bme280-digital-barometric-pressure-altitude-temperature-i2c-module	Recieved
MicroSD Card	Avionics	1	9	0	9	https://thepihut.com/products/sandisk-32gb-microsd-card	Recieved
IMU	Avionics	1	12.5	0	12.5	https://thepihut.com/products/adafruit-9-dof-imu-breakout	Recieved
LCD	Avionics	1	8.7	0	8.7	https://thepihut.com/products/waveshare-3.5inch-lcd-tft	Recieved
Battery	Avionics	2	4.6		9.2	https://hobbyking.com/en-us/turnigy-1	
ABS piping	Struct...	2	14.17		28.34	https://www.plasticpipeshop.co.uk/1-1	Not ordered
Nose cone	Struct...	1	0	0	0		Not ordered
parachute	Struct...	1	22.13	4.17	26.3	https://modelrockets.co.uk/scratch-build-parachute	Not ordered
shock cord - inextensible	Struct...	1	2	5.24	7.24	https://modelrockets.co.uk/ukroc-26-cm-shock-cord	Not ordered
shock cord - extendable	Struct...	1	4.3		4.3	https://modelrockets.co.uk/scratch-build-shock-cord	Not ordered
eyebolts	Struct...	1	4.29		4.29	https://www.amazon.co.uk/PATIKIL-M4-2mm-eyebolt	Not ordered
spray paint	Misc	1	9.99		9.99	https://www.amazon.co.uk/Fluorescent-spray-paint	Not ordered
rail guides	Struct...	1	8.1		8.1	https://modelrockets.co.uk/scratch-build-rail-guides	Not ordered
epoxy glue	Struct...	1	20.99		20.99	https://modelrockets.co.uk/ukroc-26-cm-epoxy-glue	Not ordered
epoxy resin	Struct...	1			0		Not ordered

Timeline

Avionics

Phase	TASK TITLE	TASK LEAD	START DATE	DUE DATE	DURATION (DAYS)	PCT OF TASK COMPLETE	PHASE ONE													
							WEEK 1							WEEK 2						
							M	T	W	T	F	S	S	M	T	W	T	F	S	S
1	Procurement and Initial Code																			
1.1	All Components Procured	H&P	08/12/25	10/12/25	2	100														
1.1.1	Core Sensor System Functional	H&P, S	10/12/25	10/12/25	2	100														
1.2	High-Rate Logging Achieved	S	11/12/25	12/12/25	2	100														
1.3	Data Pipeline Operational	S, A&V	12/12/25	26/12/25	7	100														
2	Power, Analysis & Design																			
2.05	All Components Procured	H&P	9/2/26	11/2/26	2	100														
2.1	Voltage Monitoring Circuit	H&P	11/2/26	13/2/26	2	0														
2.2	Critical Power Supply Tested	H&P	11/2/26	13/2/26	2	0														
2.4	Full Analysis Script Complete	A&V, S	9/2/26	23/2/26	14	0														
2.5	GPS???		TBC	TBC	TBC	0														
2.6	Flight Ready Schematic Finalized	H&P, M&D	16/2/26	18/2/26	2	20														
3	Manufacturing and Integration																			
3.1	Avionics Board Functional	All Team	TBC	TBC																
3.2	Avionics Bay Complete	M&D	TBC	TBC																
3.2.1	Full Mission Simulation	All Team	TBC	TBC																
3.2.2	Launch Readiness	All Team	TBC	TBC																

The avionics timeline has successfully transitioned from Phase 1 (Procurement and Initial Code), where 100% completion was achieved for high-rate logging and data pipelines by late December, to Phase 2 (Power, Analysis & Design). Currently, most hardware

components are procured, our breadboard prototype is functioning and the project is focused on Power supply, voltage regulation and the critical 3.3V/1.5A Power Supply for the committee payload. While the KiCad Schematic and PCB Layout are currently in progress to meet the February 21st CDR deadline, the project remains aligned with the overall Gantt chart due to the early completion of software milestones, ensuring a sufficient buffer for the upcoming Manufacturing and Mechanical Integration phases in March.

Structures

Stage	Task	Month	Completion
1	Discussing initial design concepts	October	Completed
2	Deciding the materials	December	Completed
3	Finish rocket design	January	Completed
4	Buying the materials	February	Not completed
6	Testing different parts of the rocket	February/March	Started
7	Refine rocket design and materials	March	Not completed
8	Final construction	March-May	Not completed

The structure's timeline is on track, after receiving the feedback from the PDR there have been many aspects changed in the OpenRocket design and therefore the CAD model in response to this and the design is now finalised. The next step, this month, is to start buying the materials needed such as the ABS and plywood as well as testing the forces that will be acting on the rocket. The testing will start with the nose cone being 3D printed and finding out whether it can withstand the impact force. After purchasing the materials, the other elements such as the fin roots, the shock chord will then be tested. After this the design will be refined depending on the results of the testing and the final construction will then take place.

Design breakdown

Launch vehicle design

The rocket has been designed with strength and structural integrity in mind. We decided to opt for a high power motor in order to challenge ourselves, as the higher the motor power rating, the more structurally sound the rocket has to be due to the greater forces the motor will exert.

In order to accommodate this, we researched extensively into what different kinds of materials could be used for all aspects of the rocket. We researched wood, plastics, cardboards and composites for each component to see which one had the properties that

best suited our criteria. Those criteria were to be as lightweight as possible whilst being as strong as possible.

For the main body of the rocket, we have decided to use Class C ABS piping with an outer diameter of 48.3 mm, an inner diameter of 42.7mm and has a wall thickness of 2.8mm. We are aware that less than 3mm is not ideal for the structural integrity of the rocket, especially with such a high powered motor, however we opted for this one because it was the thickest and strongest piping we could get that would allow us to actually fit the 40mm diameter payload inside the rocket. This was actually our second choice for tubing, as we initially intended to use Class E ABS piping which is more costly but is much thicker than the Class C, however its inner diameter was marginally too small to fit the payload inside, so we had to change it. We were also debating about using Polycarbonate for the main body tubing as well due to its very high strength, however decided against it as it was around double the price and we deemed that the performance increase compared to the price increase was not worth it.

For the nose cone, we were again, initially planning on using a Haack series ABS nose cone we had found that fit the diameter of our initial Class E tubing, however with the change to our tubing we had to rethink our decision. In the end, there was no nose cones at all, let alone Haack series ones, that fit the inner diameter of our tubing, so we decided to 3D print our nose cone, so we transferred the nose cone we had designed initially into the newly updated one to fit the new apogee requirement and then got that ready for 3D printing. However, again, due to the newer, lighter tubing our apogee got increased so we needed to redesign our nose cone from scratch to reduce our apogee, as it was too high and exceeded the required 609.5 m. We did this by changing to an ellipsoid nose cone as it is less aerodynamically efficient and so would lower our apogee. This then had to be redesigned in fusion as well to be 3D printed which posses advantages as it is considerably cheaper than buying a nose cone as every member has £10 of credit to use in the IForge so we don't need to actually spend anything for it.

For the fins, we just made them whatever shape permitted us to remain within the 1.6-2.2 stability range that the requirements state. They will be made from plywood as it is much sturdier than any of the other woods we would have available (such as balsa or cork). Whilst some woods like some types of spruce wood may have been a good option, it was considerably harder to source, especially in the dimensions we needed it to be in and far more expensive. Another major advantage of the plywood is that it is very easily accessible as it's available in the IForge and has the same cost benefits as the 3D printed nose cone.

For the shock chord, we debated using a Kevlar shock chord for our rocket, however we decided against it because it wasn't elastic enough and wouldn't stretch as much as we would need it to. Also, it was heavier and more expensive than the materials we actually used, which were a combination of 2 materials. We initially used just HMPE 12 as it was lighter and more extensible than the kevlar shock chord. However we found that it may still rip the bulkheads out by not being extensible enough, so we decided to add a rubber part to it to dampen the shock and lower the forces on the bulkheads so it doesn't wrench them out.

Lastly, for the bulkheads we just decided to use balsa as we can cut it easily into the specific shape and diameter we would need. Also, again, balsa is readily available in the IForge which reduces our costs.

Operations during flight

Once the rocket has taken off from the launch station, it should be a relatively straightforward flight assuming there are no hiccoughs in any of the systems and the weather is good. This is because the rocket will have a motor that will be active for a certain amount of time and then turn off and that should be about all of the systems that will actually affect the flight going up. The rocket should hopefully reach its maximum altitude after around 12 seconds of flight and the motor should have stopped burning shortly beforehand.

The recovery device (the parachute) will be set on a timer from when the motor is ignited and then will deploy a set time after this, after the 2 parts of the rocket have separated. We are aiming to have the parachute deploy just a few seconds after the apogee has been reached to bring the rocket down to hit the Earth with a ground speed of about 8 meters per second.

During flight an Inertial Measurements Unit (IMU) will record the acceleration and rotation in 3 axes. A BMP390 barometer will record the temperature and altitude. The altitude will be displayed on a PCB display so our apogee can immediately be verified without the need for a computer.

OpenRocket simulations

Our goals for the ARCS include reaching a maximum apogee of 610 m while attempting to keep the flight of the rocket as stable as possible as well as safe to fly. In addition to this we also wish to ensure the rocket hits the ground at a speed which means it will be recoverable.

Based on the simulations for the most recent and up to date design for the rocket which took place with a windspeed of 10 knots, the most updated rocket designs has a velocity off rod of 14.3 m/s. This is good because it was relatively high compared to other simulations run with other rocket designs. A higher velocity here will allow for the rocket to correct itself aerodynamically through the fins earlier on in the flight, minimising the effects of any sidewinds. This allows for a straighter flight which will be more efficient and help us reach the target height easier without having to adjust any calculations. It also reached an apogee of 674m. We feel like this is a suitable height due to the tendency of OpenRocket to overestimate the apogee by around 10%. This would give us an apogee of 606.6 which is very close to the target of 610.

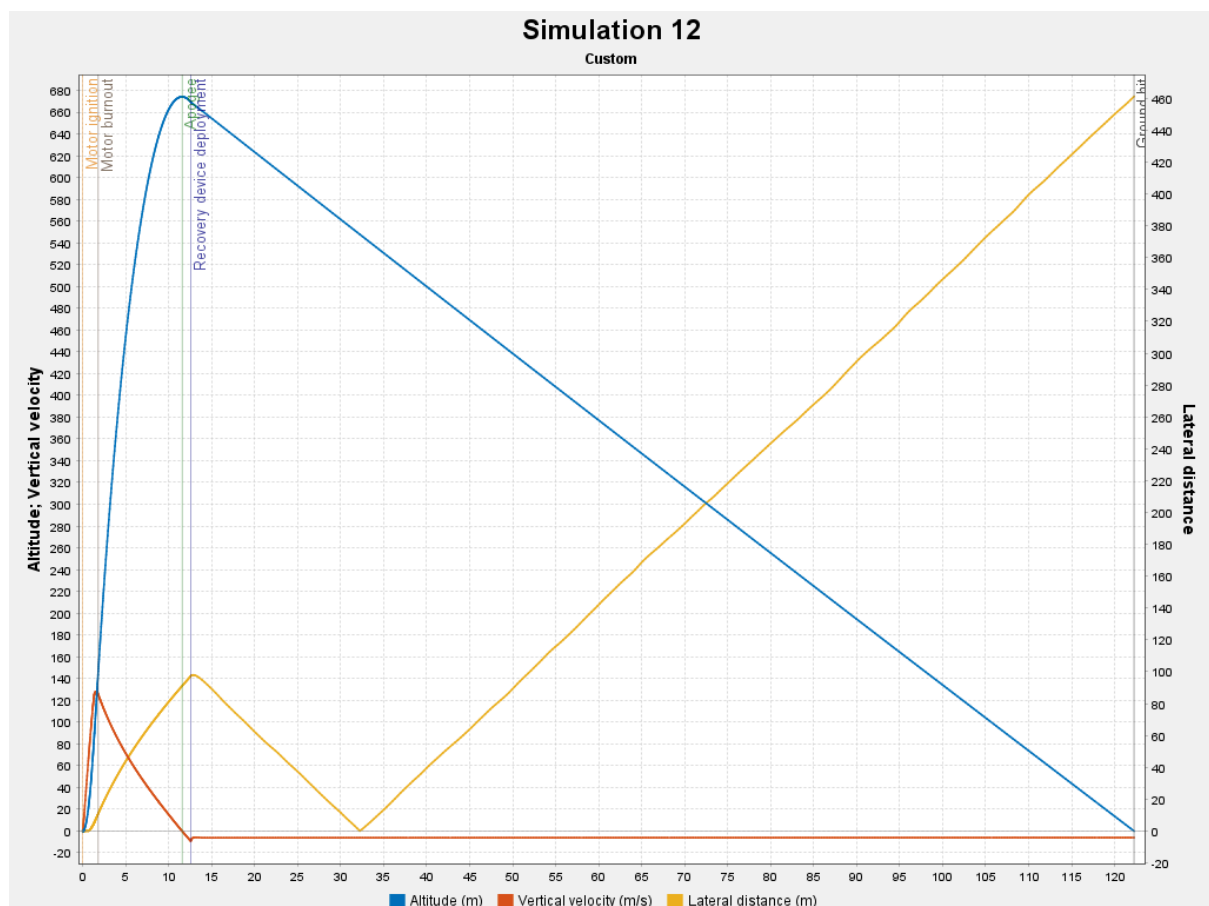
We also feel like this design is suitable as it has a stability of 2.03 calibers. We feel like this is ideal because it means the center of pressure is just over two body diameters behind the center of gravity, providing a strong but not excessive restoring force for stable flight. If stability was too much lower than this, then the rocket wouldn't have a smooth take-off as well as ascent. Especially if there is wind on the day, this would mean the rocket is prone to

being blown off course. If the rocket was too stable, it would make it prone to weathercocking so stable, steady flight would once again not be achieved.

The simulation also shows the rocket will hit the ground at 7.98m/s, which will mean the estimated impact energy is around 33J which, although remains within the toughness of ABS which is what the body and nose are made from, there are concerns about damage to the fins and the nose cone depending on how the rocket lands. We need the impact velocity to be between 7-10 m/s to help with recovery of the rocket. If it was too slow coming down, any winds could blow it miles away. Given the materials that we can find for a reasonable cost, we will have to leave the dimensions of the rocket as they are in order to ensure that all the parts of the rocket fit together without causing too much complication. As a result, we are unable to have thicker walls to protect the rocket any more on impact.

The optimum delay is the time between motor burnout, which can be seen in the figure, and the parachute ejection which occurs at the apogee. The simulation gave an optimum delay of 9.75s. This time will prevent the shock cord from tearing through the body tube so is an important safety factor

The graph shows one of the simulations run on this model. It gives an overview of the estimated vertical motion over time. It shows the motor will burn out at an altitude of around 160m. Here, the rocket will have reached a suitable velocity to reach the required apogee while balancing the forces of gravity and drag.



The graph also shows the rocket will reach a maximum lateral distance of around 460m. This would be once the rocket has landed, which is good because it won't have travelled too far away so it will be recoverable.

Recovery system

We have decided to use a laser-cut plywood bulkhead to provide a structural barrier between the recovery bay and the avionics bay. We chose plywood as it offers a good balance of stiffness and strength whilst also being easily manufacturable. The bulkhead will also carry the parachute and the attachment point for the shock cord.

Our rocket will use a nylon parachute recovery system designed to deploy on a timer shortly after apogee at a lower altitude to safely decelerate the rocket and return it to the ground with minimal damage. We chose nylon for its high tensile strength, durability and water resistance. The ejection charge in the motor will push off the nosecone and pull out the parachute. As the parachute inflates, the HMPE 12-strand shock cord will come under tension and absorb the loads between the rocket body and the parachute, ensuring a stable descent.

Once deployed, the nylon canopy will provide sufficient drag to reduce the rocket's descent to a safe landing speed.

Payload Data Analysis

The aim for the payload is to accurately record the altitude, orientation and acceleration of the rocket at all stages of its flight. This will be done by collecting several types of data and combining through MATLAB to calculate an accurate reading of the altitude and displacement of the rocket. The payload will record the current types of data:

- Altitude
- Acceleration
- Pitch, roll, yaw
- Temperature

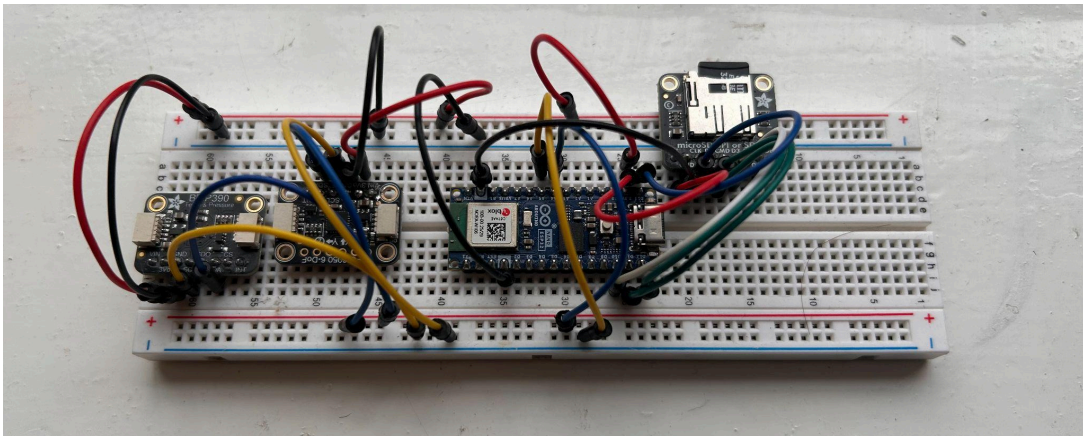
The orientation of the rocket will be presented through a simulink model where the pitch, roll and yaw data will be used as inputs to display the orientation of the rocket as it progresses through its flight. The altitude data will be presented in a simple 2D graph against time whilst the acceleration data will be shown in 3D to show the components in all 3 axes. The question the data will answer is how does temperature vary with altitude. We hope to see a strong linear relationship between the two variables. This will be presented in a 2D graph with temperature laying on the x axis and altitude on the y axis.

Data will be logged using the Arduino using a C++ program that takes the raw data from the sensors and places them into a csv file stored on the SD card. The data to be recorded will be formatted in a database by taking the csv file values from the Arduino and using a python program to add these into the database using MySQL. This is done to create an easy to use,

reusable piece of software and allows the data to be quickly and easily accessed in MATLAB, or any other programming language that may be used.

Circuit diagrams & PCB Design Overview

The prototyping phase began with a full breadboard assembly to validate the electrical compatibility of the Arduino Nano ESP32 [13] with the sensor suite. To facilitate real-time debugging without needing a laptop connection on the field, we integrated an I2C LCD Screen into the prototype to align with ARCS-SOR-3-1-3. The system utilizes a shared I2C bus to interface with a BMP390 high-precision barometric pressure sensor [14] and an MPU6050 6-axis IMU [15] (accelerometer and gyroscope). Data is recorded to a MicroSD card via the SPI protocol [16].

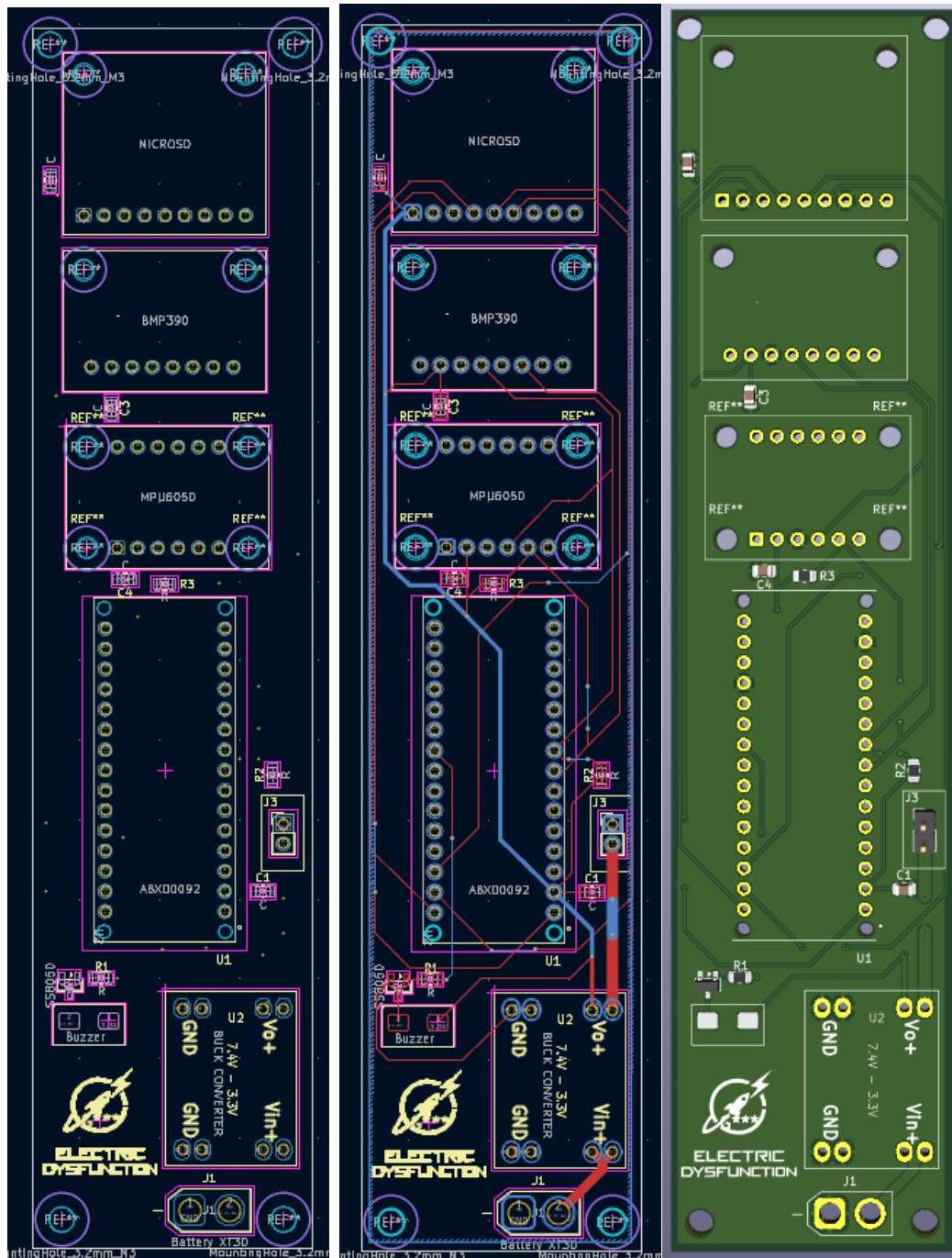


The PCB design is a custom 2-layer FR4 lead free HASL finish board designed in KiCad, featuring dedicated ground planes to shield the sensitive I2C signals from EMI generated by the battery [17]. Power is supplied by an 800mAh 2S LiPo battery 7.4V, which is regulated down to 3.3V via a high-efficiency buck regulator [18]. This is because the onboard voltage regulator on the ESP32 is too small and inefficient to handle the high currents required by the committee payload. This 3.3V is directed to the ESP32, which powers the sensor suite, data logger, and LCD screen. The rest of the current is diverted to the 1.5A 3.3V committee payload female JST-PH 2.0 connector.

All high-current traces are sized at 20mil or greater to ensure voltage stability during peak SD card write cycles [17]. The board is physically mounted to a PETG 3D-printed sled using M3 nylon standoffs, ensuring mechanical isolation from the airframe's primary structural loads.

The design includes 4.7k Ω pull up resistors and 0.1 μ F decoupling capacitors to ensure a smooth 3.3V logic level for all components[19][20]. We mitigated predicted brownouts by integrating a 10 μ F tantalum decoupling capacitor directly on the 3.3V rail [21], which has a lower ESR than standard electrolytic capacitors, allowing stored energy to be dumped faster, perfect for the SD module spikes. We originally planned to use GPS [22] to help recover the rocket, however, we ran into budget issues as we required a GPS module and multiple receivers and transmitters for the rocket and ground station [23][24], as well as concerns

about the COCOM limit bottleneck our GPS module due to high G-Force. Instead, we opted to use a loud buzzer by routing some of the 3.3V from the buck converter through a logic level transistor, rated at 2A (more than enough for the buzzer)[25].

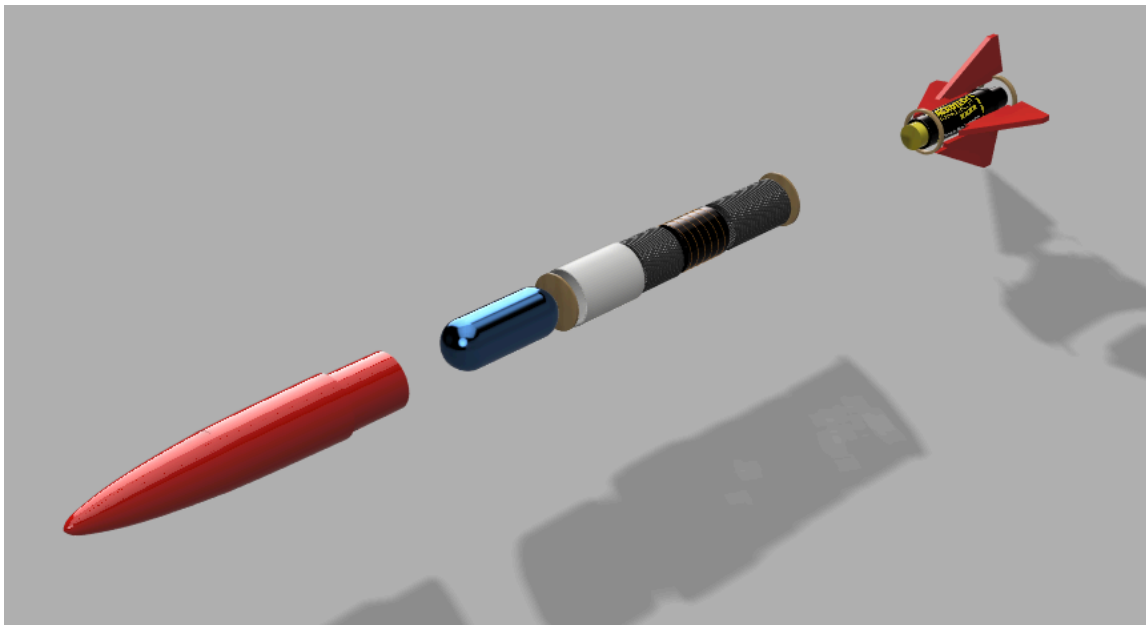
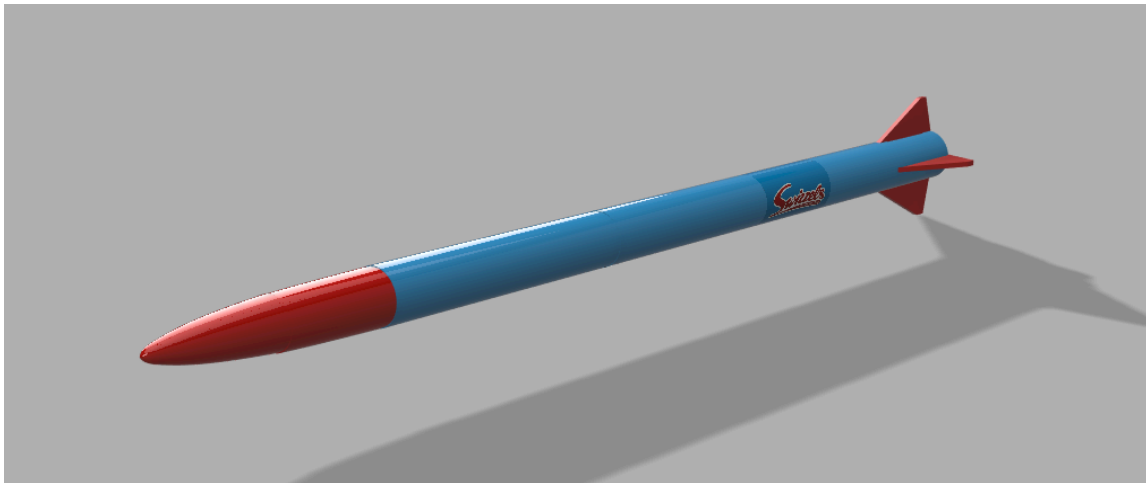


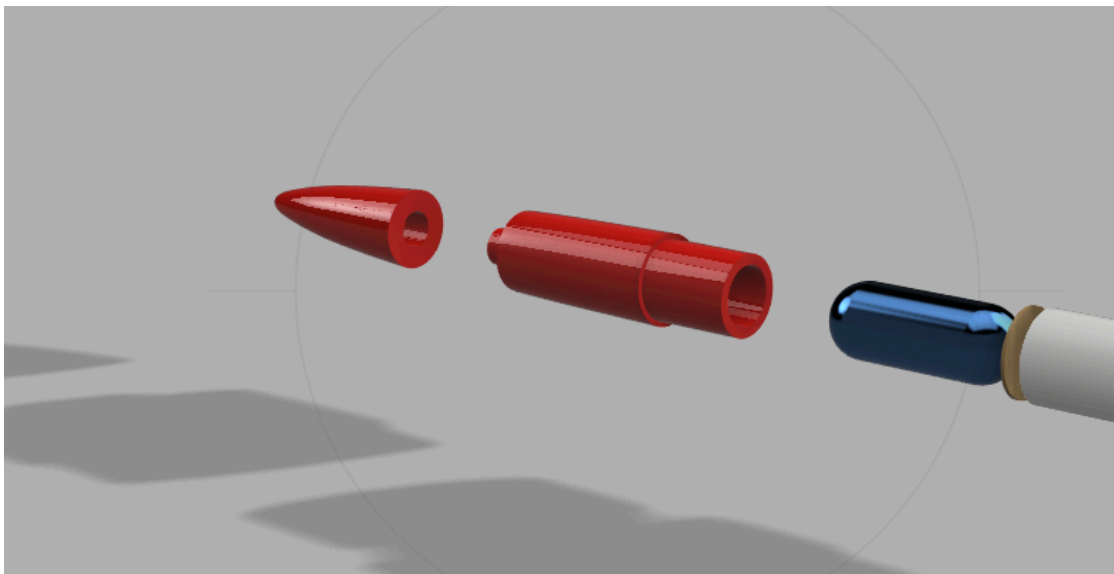
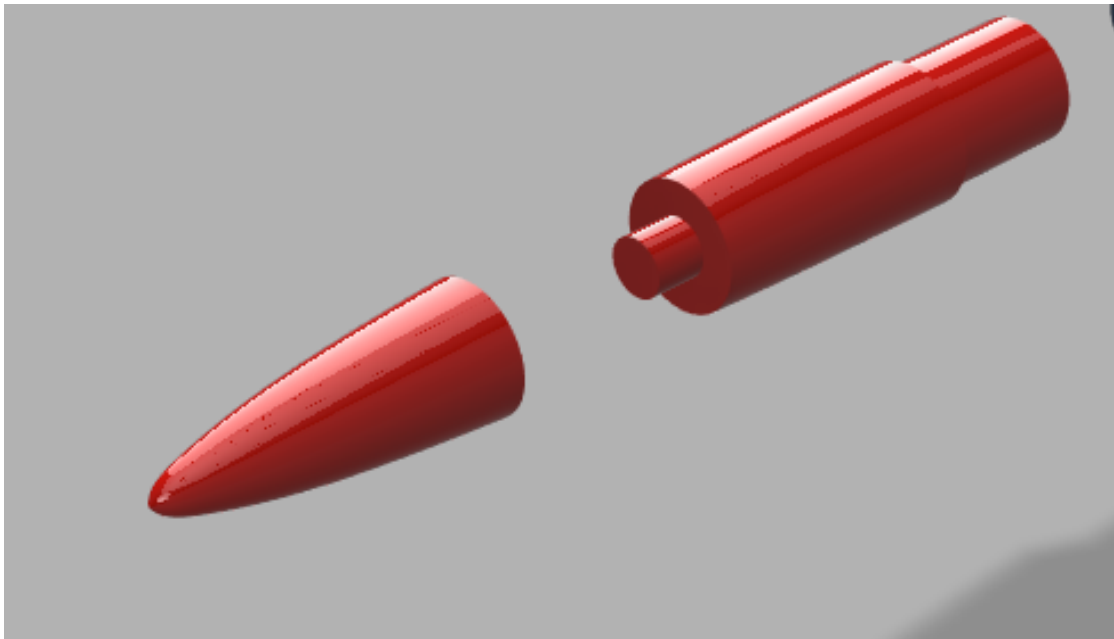
This PCB design does not include the LCD screen or the disarming switch. The positioning of these components will be decided after the release of the STL files for the committee's avionics bay, but they will be included in the final design.

While the core architecture is finalized, the PCB design remains iterative. Current development is focused on power, integrating a voltage monitoring circuit to track battery health during the 40-minute pre-launch standby. Final optimization of component density and trace routing is underway to maximize signal integrity before manufacturing the FR4 board.

CAD

Following the OpenRocket design and finalising design choices, this CAD model was then made from that design. It has gone through numerous revisions and changes as we have had to change things as we go, however this should be an accurate representation of what our rocket should look like.





Projects Risks and Mitigation

Avionics

Risk	Impact	Mitigation
I2C Bus Conflict	BMP390 and MPU6050 stop communicating.	Unique address assignment via AD0 pin. Logic level checking. Pull-up resistor

		verification.
High-G Data Corruption	SD card disconnects or stops writing during launch.	Use high-speed Class 10 cards. Implement buffered writes. SD card secured with mechanical clip.
Battery Failure/Short	Total loss of power and data, potentially frying components.	Use a smokestopper when first powering the circuit. Fuse next to LiPo. Verify all solder joints under magnification.
Vibration-induced Failure	Solder joints or headers crack under flight loads.	Use through-hole components where possible; apply hot glue to secure large modules to the PCB.

Structures

Risk	Impact	Mitigation
Shattering on impact	The rocket is then not 'reusable'.	Reduced impact speed on the lower end of the range allowed to reduce impact force on the rocket.
The rocket body failing under the stress caused by launching	The rocket will have either structural or/and functional failure.	Perform testing on each of the elements to make sure they can withstand the forces applied.
Fins bend under the drag force.	Reduces the effectiveness of the fins and stability decreases.	Using plywood so that it has a high stiffness and strength so it does not bend due to drag forces.
The rocket being under or over stable.	The rocket will not fly straight up.	Using OpenRocket which is around 90% accurate to get our stability within the range required.

Competition rules compliance

The rocket design has been heavily influenced by the Competition rules. Each design decision has been shaped around how to achieve the targets set by the rules. The following table shows how our rocket complies with each of the Competition rules.

Structures

Specification	Requirement	Compliance
ARCS-CR-1-4-1	Teams must use safe practices when building their rockets. Construction on rockets should be done using the proper equipment and spaces such as those available in the iForge	The 3D printed nosecone will be manufactured in the IForge, as well as the laser cut components such as the bulkheads and the centring rings. Assembling the rocket will require use of epoxy glue [11], so to minimise risks and danger, this will be carried out in the IForge.
ARCS-CR-1-4-2	Hazardous materials should be avoided where available. Where these materials are used they should be justified and their use should be shown to be carefully managed to prevent harm to team members	We are using epoxy glue and ABS, which can pose some health issues. However, they are less dangerous than alternative materials such as carbon fibre, and if processed in a well-ventilated area (the IForge) the risk should be minimised [7].
ARCS-CR-1-5-1	All motors for competitions must be ordered and purchased through the committee	The motor has been ordered by the committee
ARCS-CR-1-5-2	All motors and explosives must be stored following UKRA guidance.	The committee are storing the motors
ARCS-CR-1-6-1	Teams are not expected to complete test launches	No test launches are being carried out
ARCS-CR-1-6-2	Where recovery testing does not require separation testing, teams may be able to complete this testing independently, such as with mechanical systems. Please contact the committee for guidance.	No recovery testing will be needed - the charge in the rear of the motor will be used to deploy the recovery system

ARCS-CR-2-1-1	Rockets must not be designed to be intentionally unstable or understable. This will be defined within this competition of having a static margin of between 1.6 and 2.2 on the launch rail	Our stability is 2.03 cal [Appendix 4] , which is within the stability margin requirements
ARCS-CR-2-1-2	Rockets must have a length that is: no greater than 20x the width of the rocket/ no smaller than 10x the width of the rocket	Our rocket is 940mm in length, with a 48.3mm diameter. This means the rocket has a length:diameter ratio of 19.46:1 , which is within competition regulations
ARCS-CR-2-1-3	Rockets shall sag no more than 1% of their total length when suspended from both ends	This will be tested to verify.
ARCS-CR-2-1-4	Any aerosurface, such as fins, must not deflect greater than 10 degrees measured across their span (root to tip) when the total flight mass of the rocket is suspended from their tip (point of maximum distance from the body)	This will be tested to verify.
ARCS-CR-2-1-5	Motors should be mounted exclusively on the centre line and should be directed entirely through the centre line of the rocket throughout the entire flight regime	This will be ensured in our rocket by using centring rings to place the engine mount in the centre of the rocket and absorb the force applied from the thrust
ARCS-CR-2-1-6	Fins should be able to take a force equal to (2x Fin mass x Maximum G force) suspended from their tips, pulling downwards	This will be tested to verify.
ARCS-CR-2-1-7	All tube couplers must be a minimum of 2 diameters long, with each side of the connector extending a minimum of 1 diameter into its receiving body tube	Our tube coupler is 100mm in length, which is over the required 96.6mm .
ARCS-CR-2-1-8	All rockets must be equipped with 6mm rail guides. There must be a minimum of two independent rail guides that align the rocket to the rail at launch, rail guides may not be detached during flight	Two 6mm rail guides have been positioned on the body, and will be screwed on through the body to prevent them falling off during flight.
ARCS-CR-2-1-9	Rockets must not deploy components or sub-assemblies during flight. All launch	The 2 rigid launch components have been attached together by 5m of

	components must stay on the rocket. Any rocket segments that split from the rocket during recovery must be joined using a shock cord	shock cord and a nylon parachute
ARCS-CR-2-1-10	FEA is not an accepted method of proving the strength of your rocket. Any and all load testing should be done using real components under real environments.	Structural testing will be carried out on components to ensure the rocket can withstand the required loads
ARCS-CR-2-1-11	Rockets must not be designed to be carrying loose items or items which may cause harm if they are released from within the rocket. Any ballast must be secured in place such that it is immovable during all phases of flight. Additionally, materials may not be placed within the recovery bay of the rocket for "aesthetics" upon deployment. (includes glitter	All parts will be fastened to each other through glue or hooks, and there will be no glitter released upon recovery deployment. D:
ARCS-CR-2-1-12	All rockets must be designed such that they are able to be reflown (assuming that they are provided with a new motor, and if applicable recovery charges). Rockets that are not reflown will lose points in the competition	ABS was chosen for the rocket body material over cardboard due to it being stronger [7]. This increases the survivability of the rocket and its reusability.
ARCS-CR-2-4-1	Alternative methods to this require approval from the committee for use in the competition. Any recovery method that does not utilise the ejection charge of the motor as its primary recovery deployment MUST use a COTS flight computer for its recovery system.	Not necessary due to a single-chute recovery system being used

Avionics

ARCS-CR-2-2-1	All teams must ensure the place is well ventilated before soldering any component. Additionally, all teams must ensure that all solder used is lead-free.	All soldering performed in ventilated spaces using lead-free solder.
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ARCS-CR-2-2-2	Before connecting a battery to your components, verify that no short circuits are present and everything is routed correctly. We suggest using a smoke-stopper when first testing avionics. Ensure that the battery is not routed to the USB port of your device.	The battery is never routed to the USB port; electrical continuity tests are in place. A smoke stopper will be used.
ARCS-CR-2-2-3	Batteries above 7.4V are not allowed.	We will use a 2S 7.4V LiPo battery.
ARCS-CR-2-2-4	It is recommended to use multi-core wires for signal connections. Connectors such as JST and XT are recommended as well.	Use of multi-core wiring for all flexible connections to prevent fatigue. Using XT30 connector for our battery.
ARCS-CR-2-2-5	All teams must mechanically mount (i.e. with screws & bolts) their PCB; hot glue or epoxy can't be used to stick your avionics to the avionics bay. PCB mounts must be designed to resist the shock forces experienced by the rocket during its flight.	Our PCB has been designed with screw holes for mounting. Our avionics bay has been designed with dampers to reduce the effects of vibration.
ARCS-CR-2-2-6	All rockets that have electronically controlled recovery charges must have a method of externally enabling/disabling these systems via a switch and or other isolator.	The external power isolation switch is clearly labeled on the airframe.
ARCS-CR-2-3-1 - 2-3-3	Control Systems Rules	We will not use control systems in this rocket.

Statement of requirements compliance

The rocket design has been heavily influenced by the ARCS SOR requirements. Each design decision has taken into account the SOR. The following table shows how our rocket complies with each requirement.

Structures

ARCS Requirement	Description	Compliance
ARCS-SOR-4-1-1	Rockets must be reusable.	Our recovery method and

	Your rocket is expected to return in a re-flyable state. Your rocket should be designed to take all forces within its flight including impact forces without breaking.	materials used (e.g. plywood bulkhead) ensure that the rocket will only experience minimal damage, allowing it to be reused.
ARCS-SOR-4-1-2	No free moving components. Your rocket should not have insignificant unrestrained mass. All masses should be rigidly secured to the rocket. Mass components must not be able to shift during flight	All parts are securely attached to the rocket, including the fins which are slotted into the fin tabs in the ABS piping and fastened with epoxy.
ARCS-SOR-4-1-3	Thrust Forces must be directed into clear pathways. These pathways should be intentionally constructed to take motor forces. You are expected to justify your thrust force pathway.	We are using centering rings to connect the engine mount to the main rocket body, fastening it with epoxy resin. Because ABS is rigid, we don't need any structural metal components.
ARCS-SOR-4-1-4	No free falling components. All components of your rocket must return under a valid recovery method. None of your components are allowed to free-fall.	The rocket will be returned safely to the ground in one piece using the parachute, and there will be no parts separation.
ARCS-SOR-4-2-1	Rocket decoration must be appropriate. We are launching at public sites, there will be children at launch sites. Please ensure your decorations are inoffensive and preferably apolitical.	Our rocket logo is professional. We are planning on using spray paint to create a bright and colourful design on the rocket which is appropriate for anyone viewing the rocket.
ARCS-SOR-4-2-2	Rocket decoration must not significantly impact flight performance. Decorations should be cosmetic, with the exception of small aerodynamic changes such as body friction you should work to ensure that your decoration does not compromise the flight capabilities of your rocket.	Our decorations are purely cosmetic as we will be using spray paint, so this will not affect the rocket's performance.

	ARCS official's payload bay specified at 40mm in diameter	An ABS pipe with an inner diameter of 42.7mm has been selected to include this
ARCS-SOR-5-1-1	Rockets will target an altitude of 1900-2000ft and will not exceed an altitude of 2100ft	Our Openrocket Simulations predict an Apogee of 674m (2211ft) [Appendix 4] . This has been achieved through iterative design of components such as the nosecone and the fin tabs - to not exceed the Apogee we are 3D printing our nosecone to increase the weight of of the rocket
ARCS-SOR-5-1-3	Rockets are expected to fly in a stable manner	Our simulated stability is 2.03 cal [Appendix 4] , which will provide enough stability for the rocket to be stable, whilst also minimising wind cocking
ARCS-SOR-5-1-4	The flight data collected from the rocket should be inline with the simulations created by the teams for the flight conditions of the day.	The OpenRocket simulations are set to launch conditions with a 10kt launch speed, as specified in the documentation
ARCS-SOR-5-1-5	Rockets must be single stage	We have designed the rocket to only have 1 motor and still reach the desired altitude. This follows the ARCS competition rules whilst also minimising design complication
ARCS-SOR-5-2-1	Rockets must have a simulated recovery range of less than 800m from the launch site at 10kt wind speeds	From simulations of 10kt wind speeds, the simulated recovery range is 450m [Appendix 3] , which is within the maximum 800m
ARCS-SOR-5-2-2	Rockets must have a simulated impact speed of no less than 7m/s and no greater than 10m/s	Our simulated impact speed is 7.98m/s [Appendix 3] , which is within the acceptable range. The way we have achieved this is by selecting a parachute that provides the desirable amount of drag for the weight of the rocket
ARCS-SOR-5-2-6	Teams must be able to explain	We are using a 30 inch

	and justify their recovery systems throughout the project	diameter nylon parachute [3] for our recovery system. This provides a desirable amount of drag to have a suitable ground hit velocity, preventing damage to the rocket. Furthermore, nylon was selected as there is minimal friction between the material, increasing the likelihood of a successful deployment [12] .
ARCS-SOR-5-2-7	Recovery bays Must not include sharp objects, shrapnel or points which may cause harm or damage the recovery devices.	The recovery bays do not contain any sharp objects

Avionics

ARCS-SOR-2-1-1	Your payload should be a minimum of 150 grams.	We exceed this around (165g ATOW)
ARCS-SOR-2-1-2	Your payload must include a custom PCB.	Custom-designed PCB integrating all flight sensors.
ARCS-SOR-2-1-3	Your custom PCB must include a permanent imprint or etching of your team's logo/name	Shown in PCB design.
ARCS-SOR-2-1-4	Your custom PCB must include mounting holes for screws to mechanically secure the PCB to the avionics bay.	Shown in PCB design
ARCS-SOR-2-1-5	Your payload must be secured and physically mounted within the rocket.	Shown in Avionics Bay design.
ARCS-SOR-2-1-6	The avionics bays must not be pressurised. (Generally this applies to the whole rocket with the exception of the recovery bay).	The avionics bay is non-pressurized (vent holes included for BMP390).
ARCS-SOR-2-2-1	Your payload must be capable of measuring and displaying rocket apogee. This data must also be able to be shown via a graph or	Shown in Payload Data Analysis Section,

	simulation.	
ARCS-SOR-2-2-2	Your payload should be able to power itself without issue for at least 40 minutes.	Power system validated for >40 minutes of continuous operation.
ARCS-SOR-3-1-1	Your payload must be capable of recording at least 4 types of data.	Data logged at 100Hz in .csv format for post-flight analysis.
ARCS-SOR-3-1-1	Your payload must be capable of recording at least 4 types of data.	We will be recording temperature, altitude, acceleration, and orientation.
ARCS-SOR-3-1-2	Your team must have a research question related to at least one of the data points you are recording.	The question we will be answering is "how does the temperature of the atmosphere change with altitude" we will be using our temperature data to answer this.

Manufacture methods

Structures

Part	Material	Method
Tubing	ABS	Abs tubing will be bought online then cut down to the required lengths. Holes will be drilled where required for the payload
Nose Cone	PETG	The nose cone will be 3D printed with a 20% infill rather than purchased as we cannot find a nose cone available online to fit the dimensions of our body. It will be attached to the body via screws as it needs to be removable to access the avionics bay.
Fins	Plywood	The fins will be laser-cut and attached to the body tube via epoxy glue.
Bulkheads	Plywood	The bulkheads are laser-cut and attached to the body tube via epoxy glue.

Centering rings	Plywood	The centering rings will be laser-cut and attached to the body tube
Coupler	PLA	3D printed
Inner tube	Cardboard	
Shock cord	Elastic cord	Attached to the bulkheads via metal eyelets.
Rail Button	Delrin	Purchased and attached to boxy with epoxy glue

Avionics

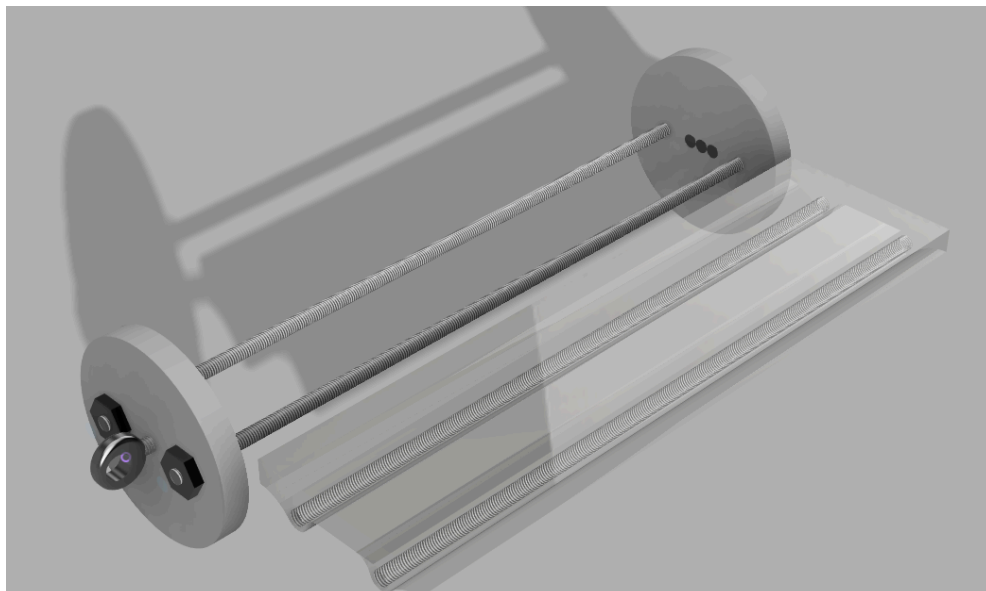
PCB

Once the PCB is finalised, and the pull up resistors, decoupling capacitors, and transistor has been soldered onto the board, it will be shipped to us, where we will solder all our sensors, our SD module, battery connectors, buzzer and microcontrollers will be mounted and soldered in their correct positions. Everything is hand soldered using flux core lead free solder[26]. Post soldering, the board is cleaned with IPA to remove residual flux, which can become conductive in high humidity launch conditions[27].

Avionics Bay

The avionics suite will be housed on a 135x38.5mm single piece of 3D printed PETG, selected for its thermal stability [28] to withstand pre-launch standby periods (ARCS-SOR-2-2-2). This sled slides along dual M3 stainless steel threaded rods, which act as a structural backbone to absorb longitudinal tension during parachute deployment. The assembly is capped by 42mm birch plywood bulkheads, secured with M3 Nyloc nuts and shoulder washers to resist high-frequency motor vibrations.

To ensure data integrity under load, the PCB will be mounted on M3 nylon standoffs, while heavier components like the 800mAh battery are centered to reduce radial forces. Neoprene foam and zip ties provide secondary vibration dampening for modular sensors. The airframe will feature three 120° offset 2mm static ports [29], allowing the BMP390 to sample ambient pressure without bay pressurization (ARCS-SOR-2-1-7). Electrical safety is maintained by an arming switch mounted perpendicular to the axis of travel.



Testing

Avionics

The following 3 stage testing protocol will be implemented upon completion of the PCB Assembly. The tests are designed to validate data logging, and mitigate risks associated with the compactness of the AV bay and the high vibration environment.

Stage 1 - Electrical Continuity Test - Before the initial power-up, a Point-to-Point Continuity Test will be performed using a digital multimeter, specifically focusing on the tight tolerances around the transistor and buck converter as well as ensuring the M3 threaded rods are isolated. Finally, we will power the PCBA on via a smoke stopper, to ensure there are no shorts.

Stage 2 - Power and Thermal Testing - To satisfy ARCS-SOR-2-2-2, the system will undergo a 60-minute Endurance Test to ensure we can power the circuit for longer than 40 minutes. The avionics bay will be fully assembled and powered by the 7.4V LiPo. We will monitor the internal temperature of the enclosed tube to ensure that heat dissipation from the buck converter does not melt the PETG sled or cause thermal drift in the BMP390 altitude readings.

Stage - 3 - Simulated Flight and Environmental Stress Test - A vacuum chamber simulation will be used to validate the altimeter's logic. We will reduce ambient pressure to simulate an ascent to the target apogee, verifying that our software correctly identifies "Launch", "Apogee", and "Landing" states. Finally, the assembly will be subjected to a mechanical vibration test, simulating motor thrust while actively logging data at 100Hz. We will then analyse the .csv file for any data skips or corruption, as well as confirm the structural rigidity of the PCBA.[30]

Structures

Initial standard tensile testing will be done on the Class C ABS piping, using the Zwick Proline Z020 stress testing machines in the diamond. As we had to change the wall size to 2.8mm, we decided it was important to test if the material would still be strong enough to withstand recovery.

Once the rocket is fully assembled, we will suspend the rocket at both ends and test how much it sags, by measuring the difference in sagging between the rocket being supported at the centre of mass and with no support except for the two at each end. This is to ensure that it sags less than 1% of its total length in accordance with ARCS-CR-2-1-3. Furthermore, we will suspend the rocket with strings attached to the tip of each fin, and then use a protractor to measure the angle of deflection of the fins from the horizontal. This is to ensure that the angle is under 10 degrees as required by ARCS-CR-2-1-4. Finally, the rocket will be suspended sideways and have a hanging weight equal to double the fin mass tied to the tip of a fin. This will be tested with the fin facing both outwards horizontally and downwards.

This is to ensure the rocket meets ARCS-CR-2-1-6, with the fins being able to withstand a force equivalent to double its weight.

We also elected to do drop testing with the parachute by attaching a weight to it and dropping it from a set height, using slow-motion filming to measure the time taken to fall. This would let us calculate how much drag the parachute provides and get an experimental estimate for recovery speed. Furthermore, we decided to drop test our 3D printed nose cone to ensure it would withstand recovery. This will be done by adding weight to the nosecone to simulate the rest of the rocket and dropping it from a height that will provide a greater impact speed than on the launch day.

Design changes from PDR

- Changed to using PETG instead of polycarbonate/ABS as we have decided to 3D print the nosecone instead of buying it from a supplier.
- Changed from using the Haack series (slide 10 on PDR) nosecone to ellipsoid as it reduces the apogee to meet the requirement and is also structurally stronger and so less likely to snap on impact, which is a bonus.
- Length of the nose cone went from 254mm to 240mm to make it easier to 3D print.
- Base diameter of the nose cone went from 48.5mm to 48.3mm to suit the new ABS pipe diameter.
- The wall thickness of the pipe went from 4mm to 5.5mm.
- The ABS pipe thickness has been reduced to 2.8mm to accommodate the avionics.
- The ABS outer diameter has been changed to 48.3mm and 42.7mm inner diameter as it is easier to find stock ABS pipes of this size and also would allow us more freedom when fitting the avionics.
- Increased the thickness to 3mm to align with the Iforge stock thicknesses.
- In the PDR, material choice was split between Polycarbonate and ABS in PDR but ended up going with ABS because of the good strength to weight ratio and the fact that it is much cheaper than the alternative.
- Added the ARCS supplied payload (based the weight off our payload) to the open rocket design to help with simulation accuracy.

Issues Encountered and Lessons Learnt

Structures

One issue we encountered was that our rocket's apogee was predicted to exceed the maximum value specified. To prevent this we redesigned the nosecone, changing from a Haack Series to an ellipsoid shape to reduce aerodynamic efficiency and therefore apogee as well. We were also forced to change our tubing from class E ABS to class C ABS due to the introduction of the 40mm diameter payload which would not fit in our original choice. Another issue in one of our initial designs was the centering rings interfering with the fin tabs,

however we were able to rectify this by redesigning a model with appropriate spacing of the centering rings on either side of the tabs. We also did not initially consider how the hot gas from the charge on the motor would reach the upper bulkhead. This will be solved by putting holes in the lower bulkhead allowing the gas to pass through. All of these issues required further simulations to be carried out and the creation of new models. One issue we continue to grapple with is that the majority of model rocketry components need to be shipped from America which greatly increases our costs. The team is attempting to be as prudent with our budget and deliberate with our component selection as possible to manage this issue. These issues have reminded the team of the importance of having a full understanding of all the requirements before beginning design but also the importance of continuing to remain as flexible as possible to combat the inevitable problems we will face later in the project.

Avionics

During the development of the avionics suite, our team resolved several critical challenges as a result of the compact 43mm airframe and the rigorous ARCS competition requirements. Initially, we encountered predicted brownouts caused by high-current transients during Micro SD card write cycles, which would temporarily starve the I2C sensors of power. We mitigated this by integrating a 10 μ F tantalum decoupling capacitor directly on the 3.3V rail and widening the main power traces to 30-mil to ensure a stable supply. Furthermore, the tight 2.54mm pitch of our buck converter triggered Design Rule Check (DRC) clearance violations of under 0.2mm, posing a short-circuit risk during manual soldering. We resolved this by redesigning the footprint with oval pads, increasing the pad-to-pad clearance to a safe 0.3mm. Furthermore, because verified libraries for our specific SD and GPS modules did not exist, we had to meticulously build custom footprints from scratch using datasheet tolerances and digital calipers. We will validate these designs by test-fitting the physical components onto a 1:1 scale paper printout of the board before ordering the FR4, eliminating the risk of manufacturing delays. Signal integrity was also addressed. To combat parasitic capacitance and EMI degrading our I2C communication, we implemented a continuous bottom ground plane and added physical 4.7k Ω pull-up resistors to the SDA and SCL lines, securing the mandatory 100Hz logging rate (ARCS-SOR-3-1-1). Finally, to manage thermal buildup inside the sealed tube during the 40-minute pre-launch standby (ARCS-SOR-2-2-2), we utilized a high-efficiency switching buck converter instead of a linear regulator and mounted the assembly on a heat-resistant PETG sled (75°C), guaranteeing the hardware will not overheat or deform on the launch rail.

Appendices

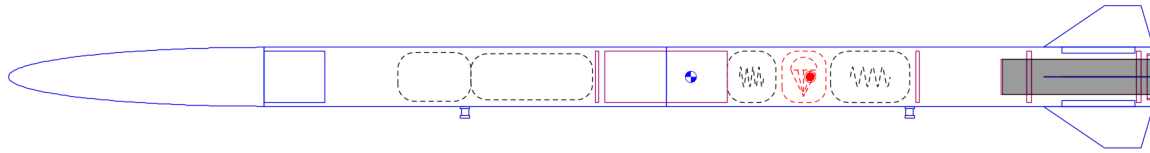
motor	Velocity off rod (m/s)	Apogee (m)	Velocity at deploy ment	Max velocity (m/s)	Max acceler ation (G)	Time to apogee (s)	Flight time (s)	Ground hit velocity (m/s)
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			(m/s)					
G78-7	14.3	674	11.1	128	10.8	11.6	122	7.98

Appendix 1: OpenRocket Simulation values

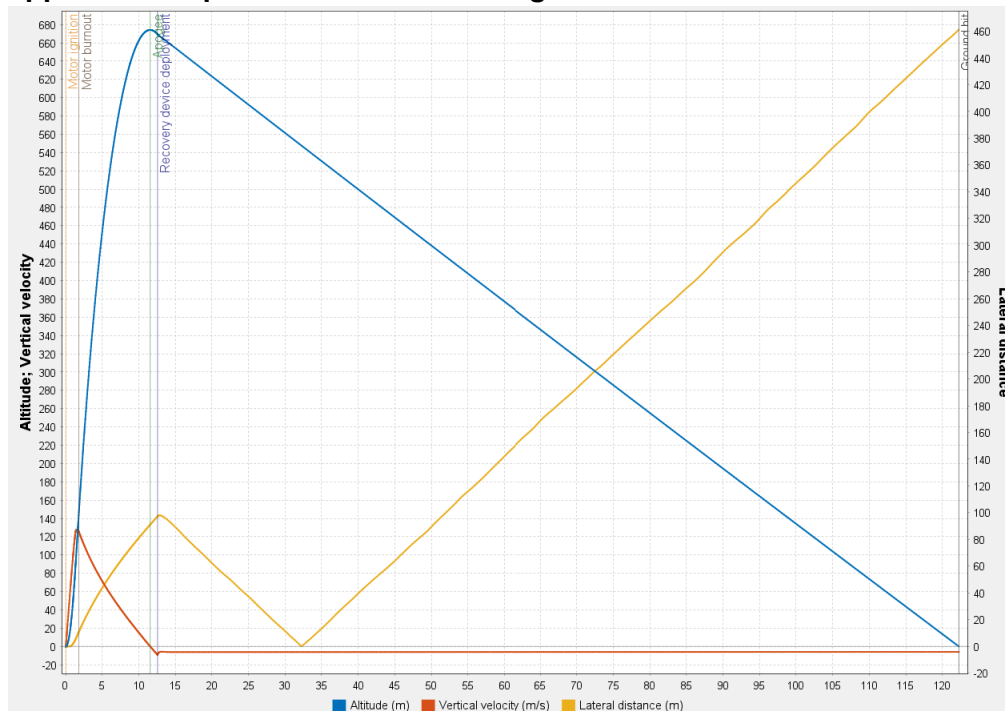
SSI Rocket CDR NEWWWWWW USE THIS ONE PLEASSSEEE
Length 940 mm, max. diameter 48.3 mm
Mass with no motors 760 g
Mass with motors 888 g

Stability: 2.03 cal / 10.4 %
CG: 560 mm
CP: 658 mm
at M=0.300



Flight configuration: [G78-7]
Apogee: 675 m
Max. velocity: 129 m/s (Mach 0.380)
Max. acceleration: 10.8 G

Appendix 2: OpenRocket Rocket Design



Appendix 3: OpenRocket 10kt wind simulations

SSI Rocket CDR NEWWWWWW USE THIS ONE Stability: 2.03 cal / 10.4 %
Length 940 mm, max. diameter 48.3 mm
Mass with no motors 760 g
Mass with motors 888 g

CG: 560 mm
CP: 658 mm
at M=0.300

Flight configuration: [G78-7]
Apogee: 675 m
Max. velocity: 129 m/s (Mach 0.380)
Max. acceleration: 10.8 G

Appendix 4: OpenRocket statistics

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[ul7fT6ta-OwGleN1IOXesSYPdLoT9D_tc&dib_tag=se&keywords=spray+paint&qid=1769168290&sprefix=spray+paint%2Caps%2C143&sr=8-8](https://www.modelrockets.co.uk/forums/threads/loc-precision-rail-guides-1000-6-pack.1769168) (accessed Feb. 17, 2026).

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